

Visualizing Lattice-based Post-Quantum Cryptographic Algorithms

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Cryptography · Post-Quantum · Another keyword.

1 Introduction

Cryptography is the study of techniques and protocols relating to securing communications (Rivest, 1990). Typically, this consists of constructing and analyzing protocols that prevent outside parties from reading private messages.

Prior to the modern era, most of cryptography was encryption, which is a process that transforms some form of information (plaintext) into an alternative form that only certain parties can decode (ciphertext). What is now known as classical cryptography consisted of using ciphers that could be practically computed and solved by hand. However, after devices such as the Enigma machine and other computational devices were developed, these ciphers became trivially easy to break.

Now, history repeats itself as the development of quantum computers threatens to render multiple modern cryptographic algorithms useless. Quantum algorithms such as the ones proposed by Peter Shor () are theorized to be able to break protocols like AES and ... given a strong enough quantum computer.

Because of this, NIST (National Institute of Standards and Technology) ran out a call for potential algorithms to

1.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

1.2 Related Works

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z$$

(1)

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

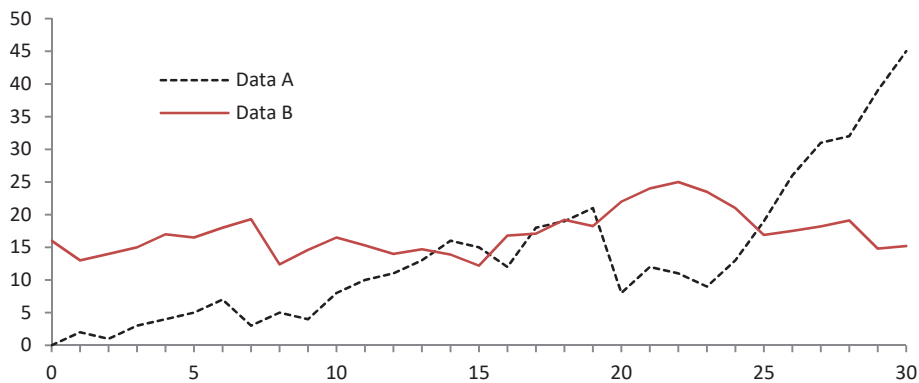


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments³, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

2 Introduction

2.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

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